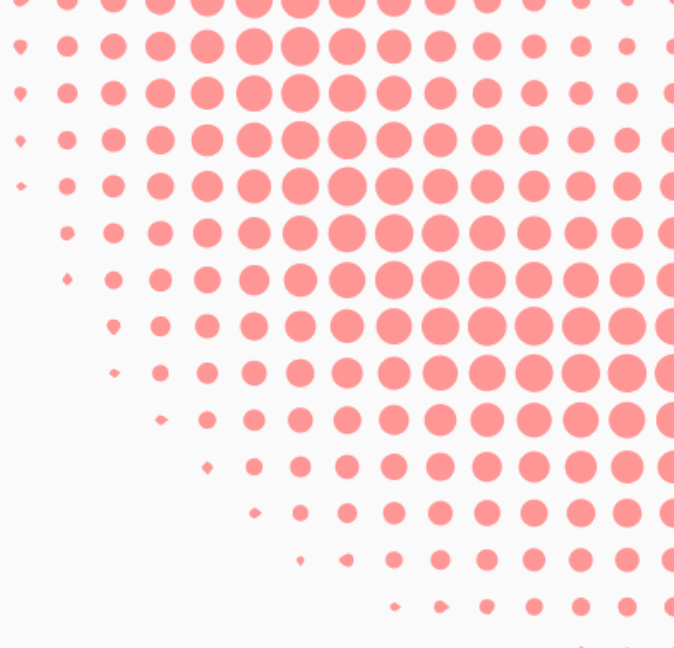




ANALYSIS OF HUMAN REACTION TO DIFFERENT MUSCLE STIFFNESS AND APPLICATION TO THE ROBOT

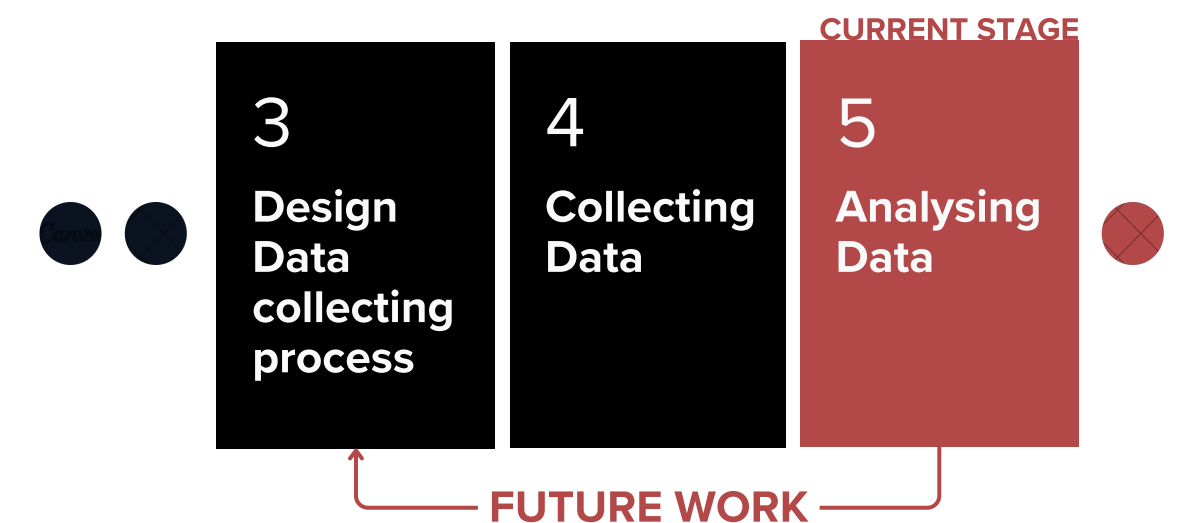
by Ben Srisoontorn

日本のプレゼンテーション



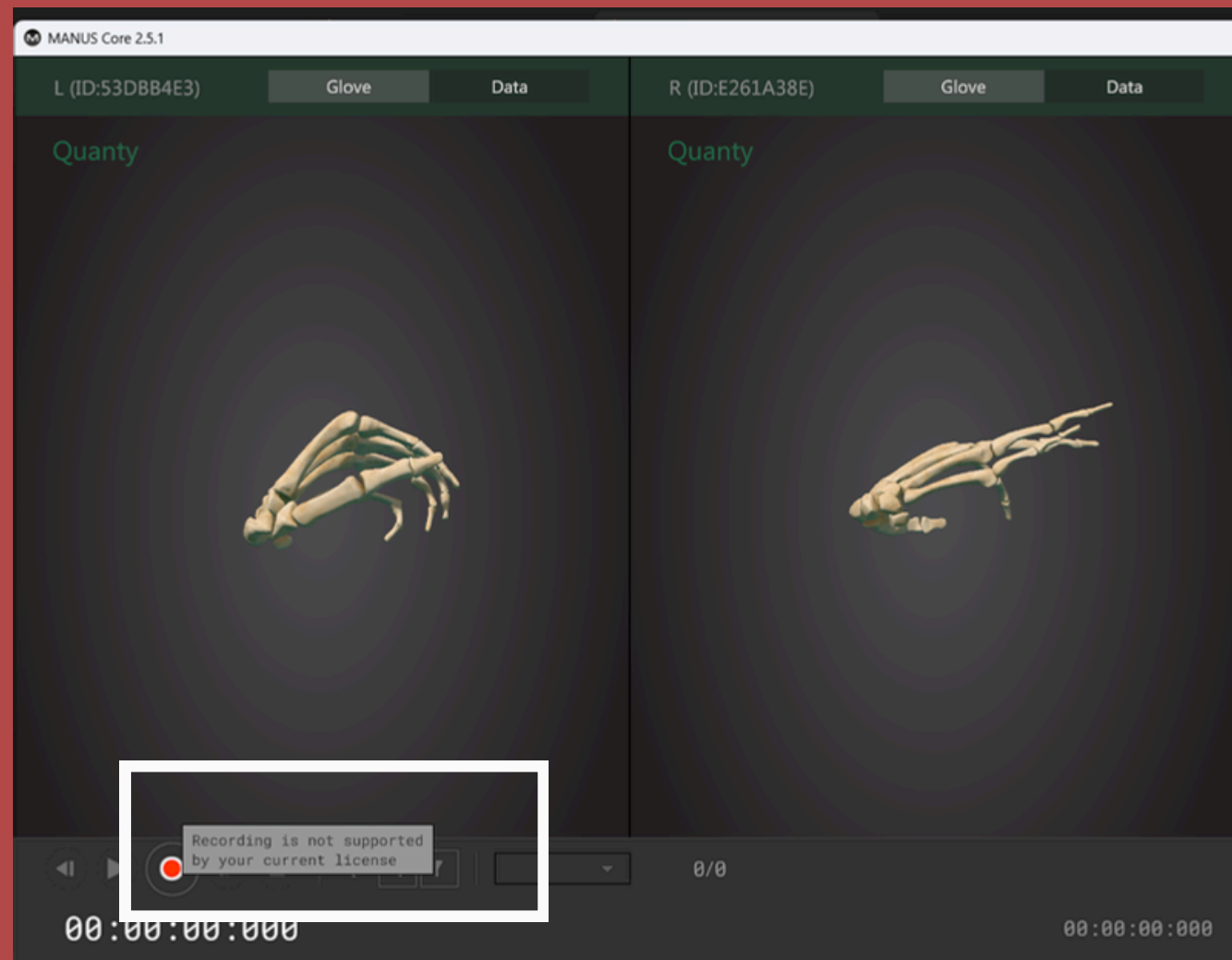
This week's progress

- Redownload the **version 2.5.1** of manus core (**glove software**)
- Plot relationship between **Force and Position** from **last week** work
- **Try connecting and reading** force sensitive resistor (FSR)
- Research **calibration method** for force sensitive resistor (FSR)
- Try **calibrating** the force sensitive resistor (FSR)



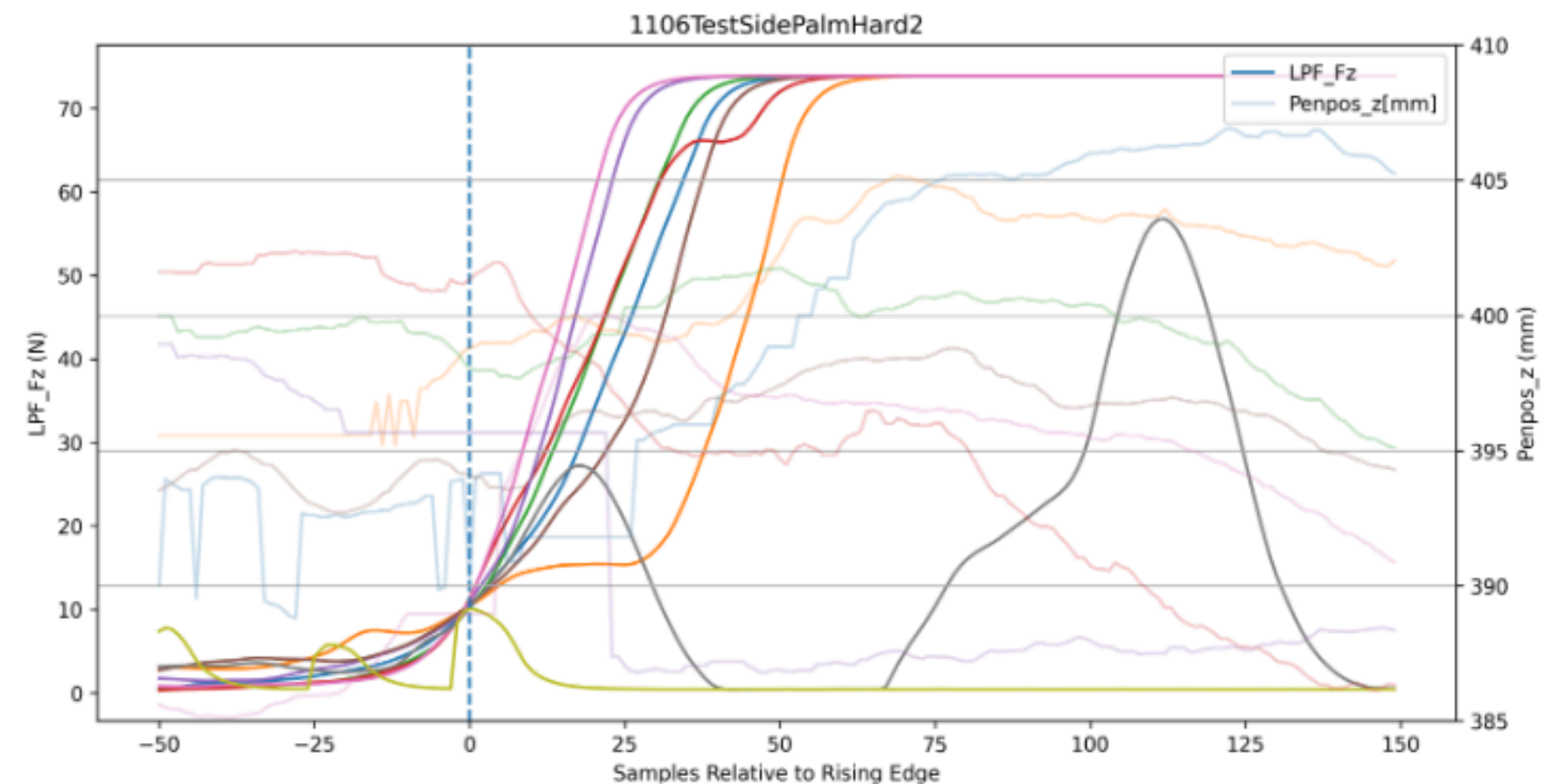
Manus core 2.5.1 install

Still need license to record



Fz - PosZ of the pressing hand

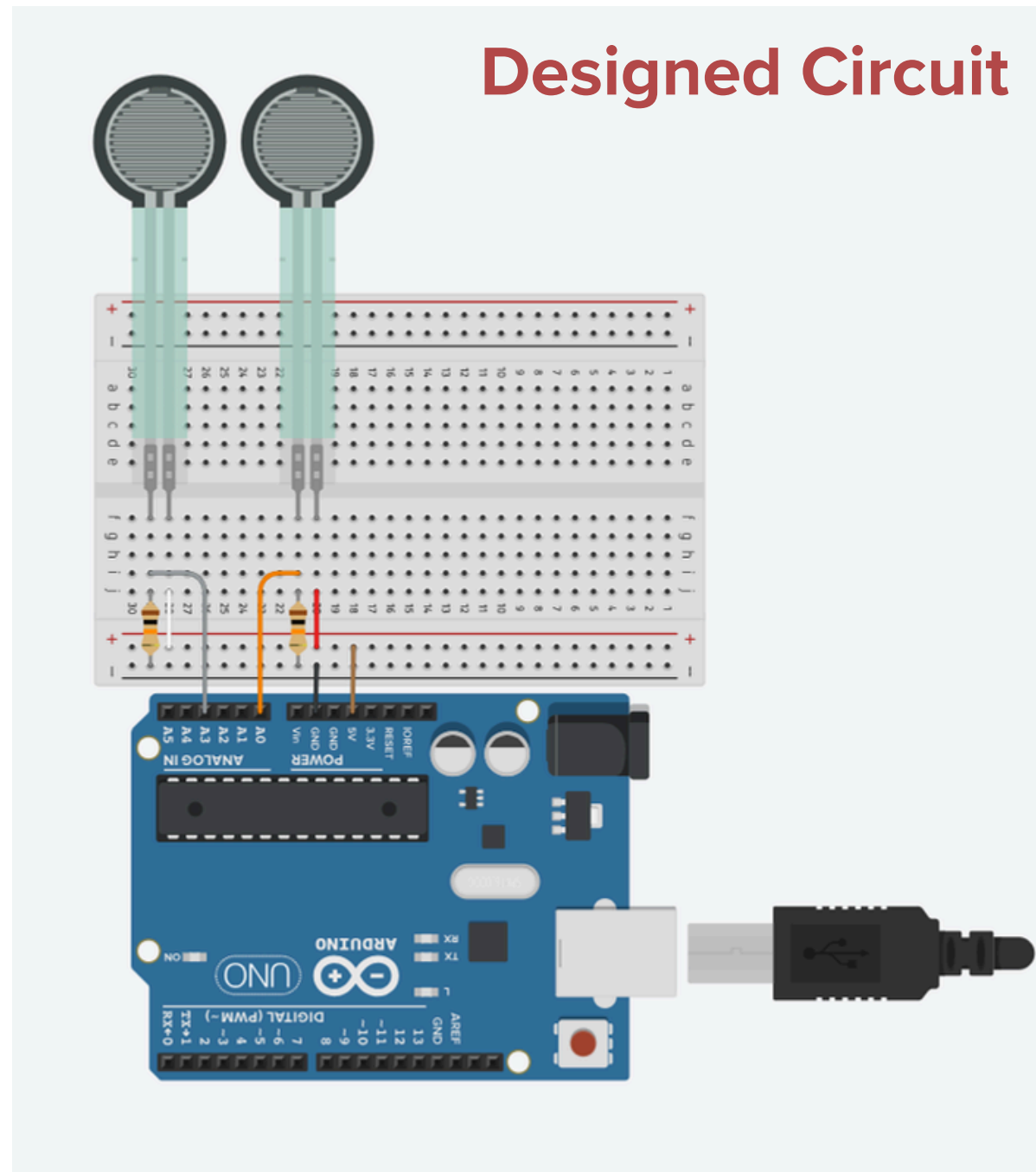
The data is too messy to analyse
Better recollect → marker info. not overall



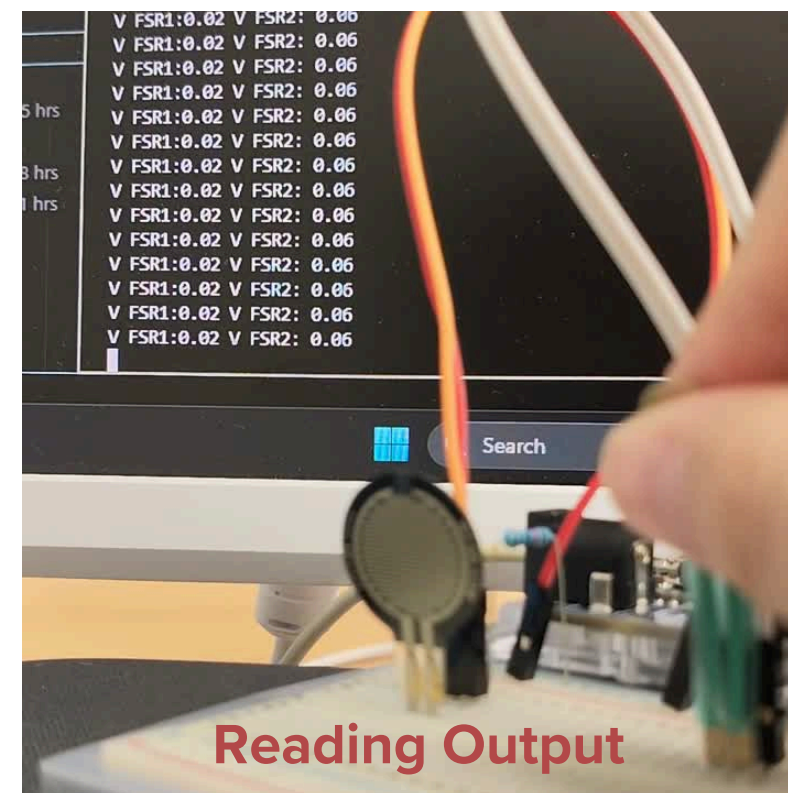
To check the designed circuit

Connect FSR to Arduino

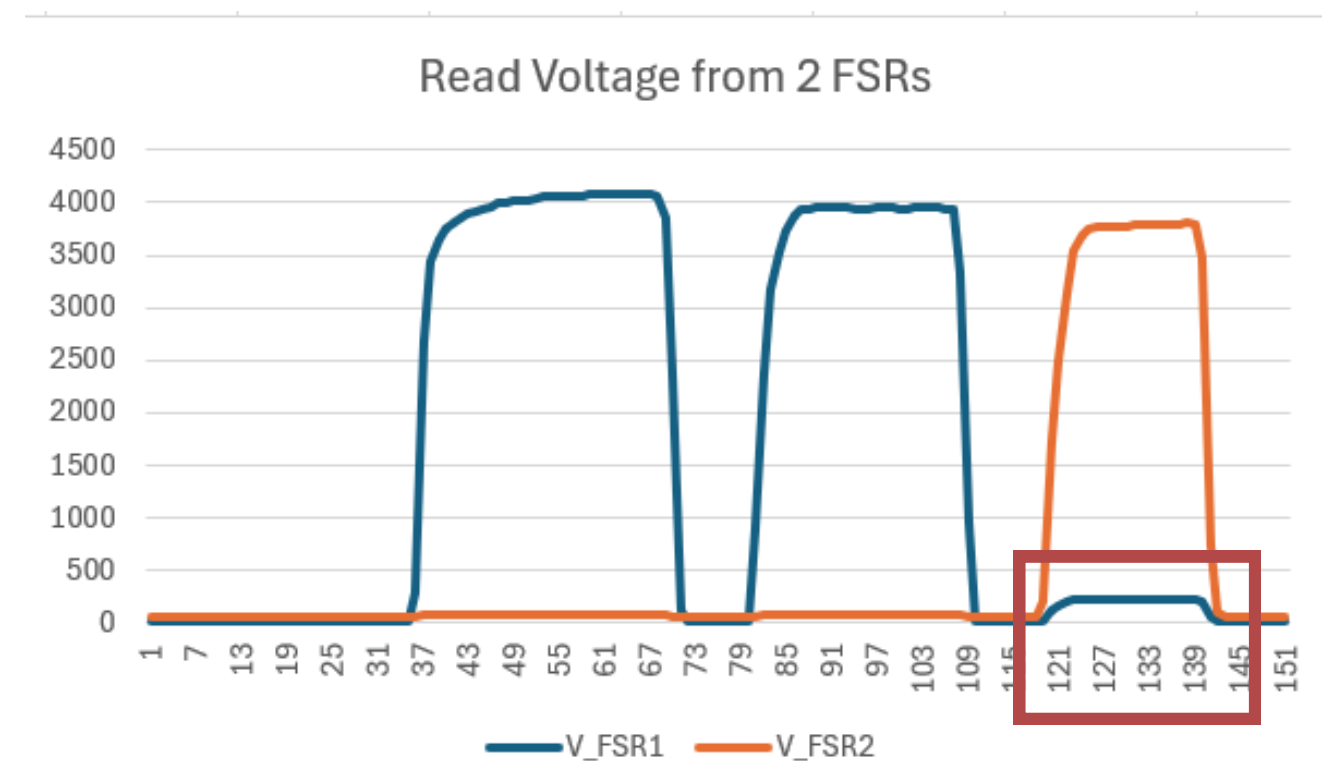
Designed Circuit



- Connect to computer and save to csv by pyserial
- Spot a noticable increase on a sensor while the other one is pressed
- Caused by breadboard? → solder and connect directly → still not changed
- Not enough current? → FSRs, resistor, don't required much current
- Better to assign threshold when sensor is pressed



Reading Output



Research For Calibration Method

Research paper 1: Calibration of force sensing resistors (fsr) for static and dynamic applications

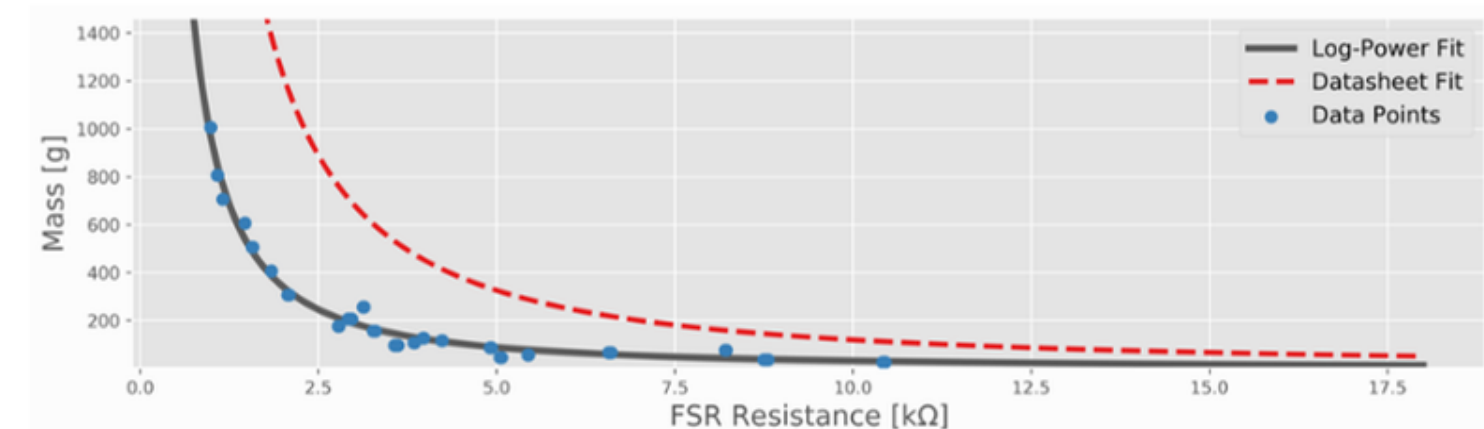
- static force → FSR give a creep of 0.2V/s
- dynamic force → Increasing force curve is not equal to decreasing force curve
- suggest using polynomial relationship with moving integral between read voltage and force

$$F(V, I) = a_0 + a_1 V + a_2 V^2 + a_3 V^3 + a_4 V^4 + a_5 I + a_6 I^2 + a_7 I^3 + a_8 I^4 \quad (1)$$

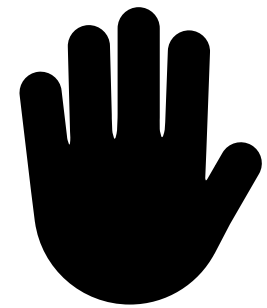
Article 2: Force Sensitive Resistors (FSRs) with Arduino

- Use reference mass to calibration the sensor
- Resistance of FSR and applied force is in logarithmic relationship.

$$\log_{10}(m_i) = a \cdot \log_{10}(R_{FSR}) + b$$



Collected Data



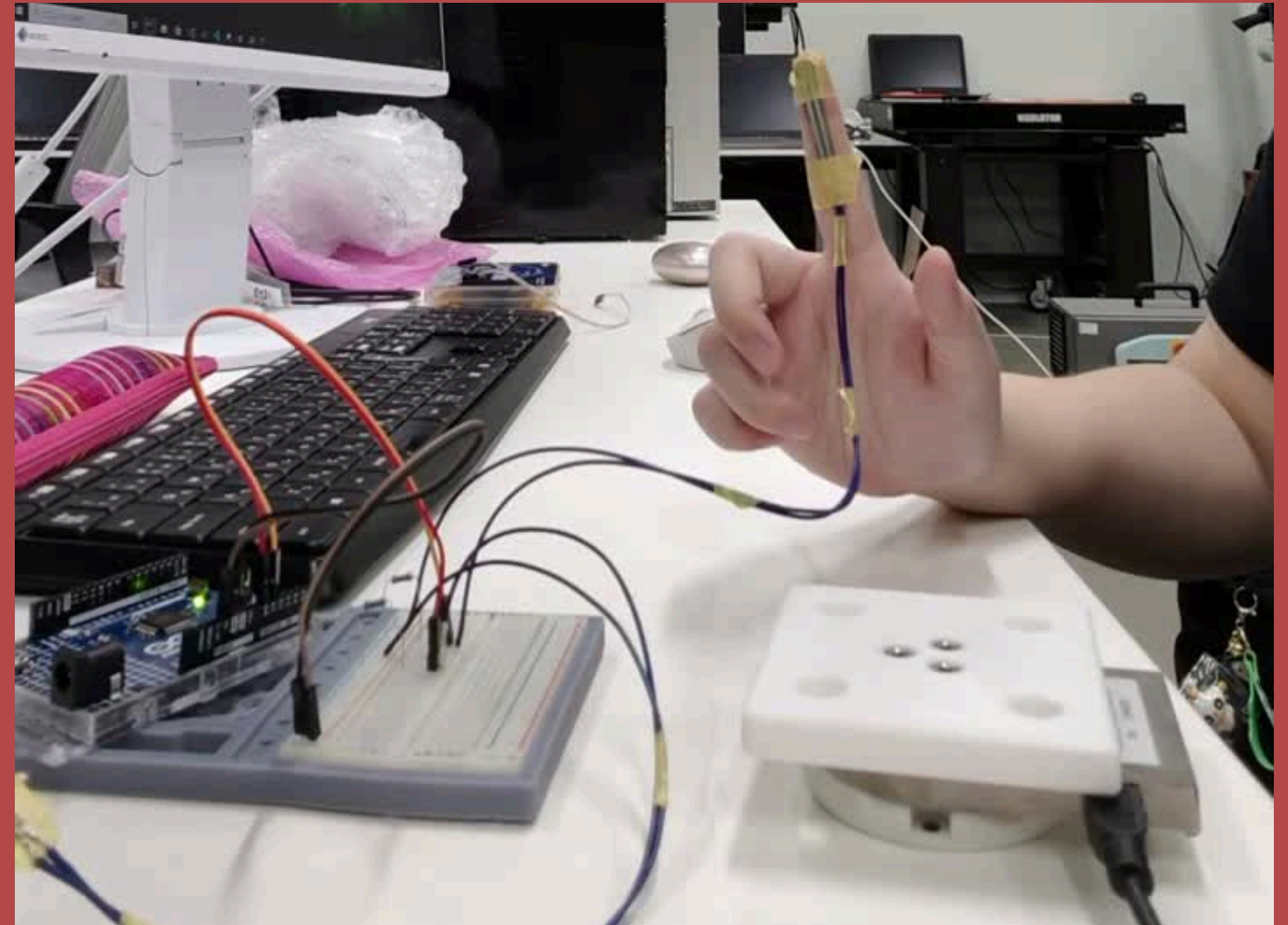
Voltage of FSR on hand
read from Arduino:

Volt_FSR1, Volt_FSR2

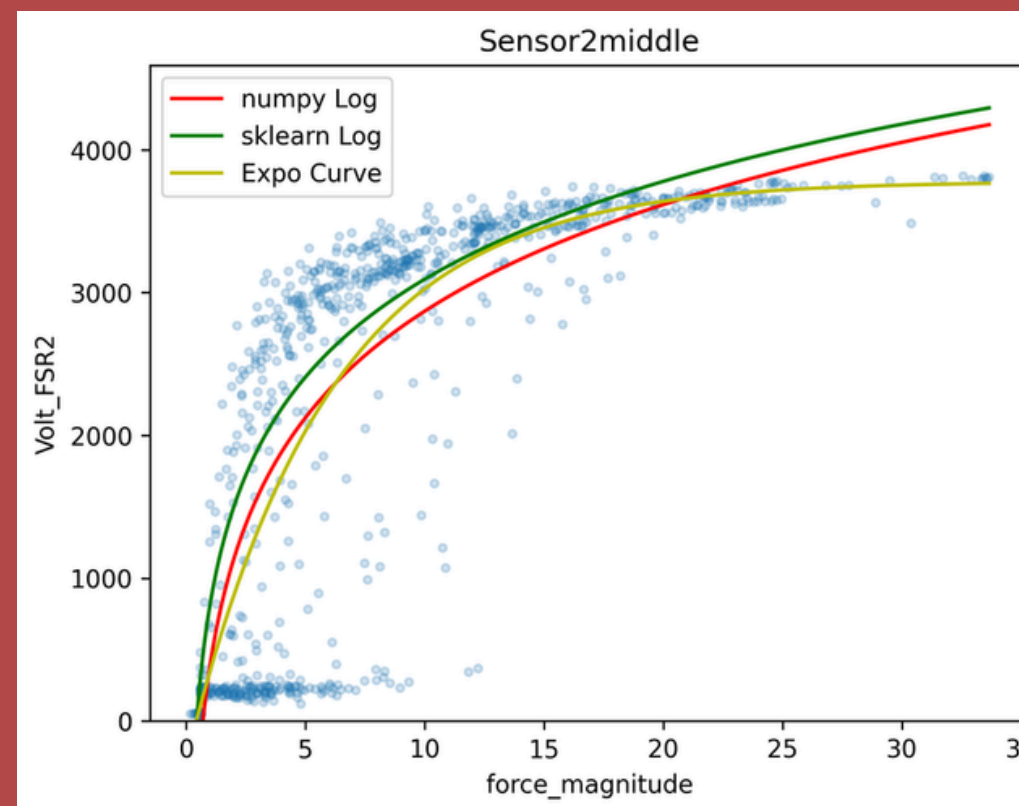
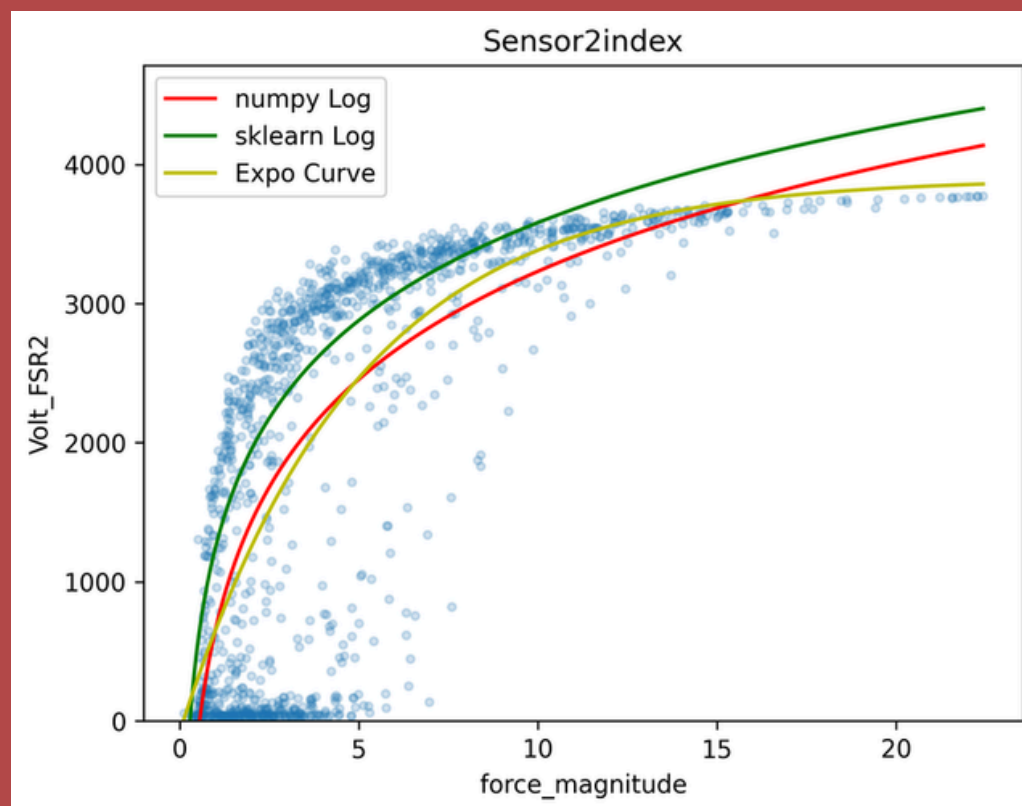
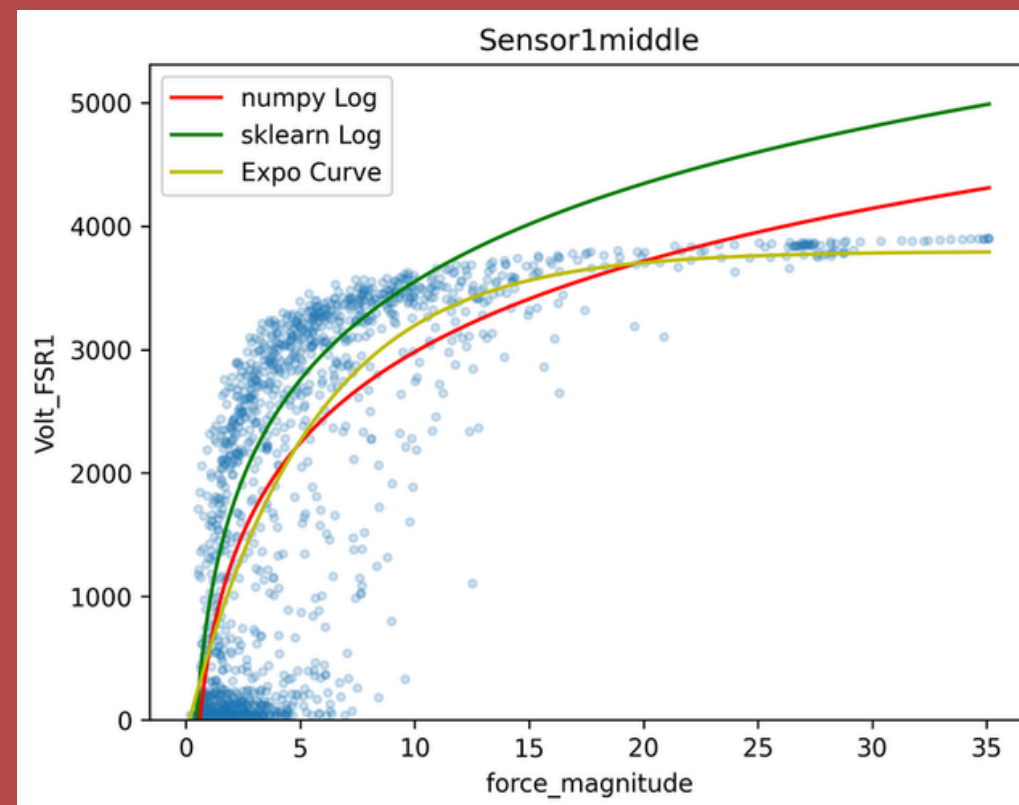
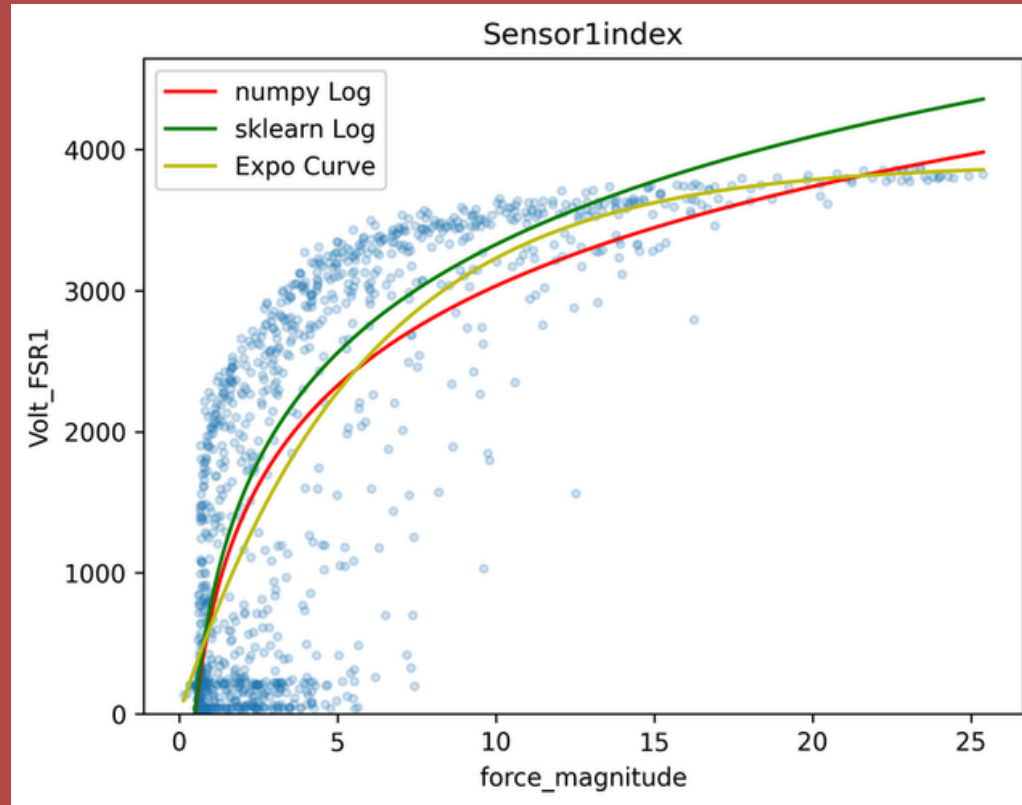


Pressed force read by
Leptrino force sensor:

Force_magnitude



Scatter Plot between Read Voltage and Force with various types of Regression



Used regression in plot

- Logarithmic regression by numpy library (red line)
- Logarithmic regression ignoring outlier by sklearn library (green line)
- Exponential regression saturation model (yellow)

Spotted problem

- Many false data points at the bottom affect regression too much
- No regression that perfectly fit the majority of the plot
- Cannot see whether the model is good or not by scatter plot

Future Work

- Apply **touching threshold** to the data and **Redo the regression**
- Plot the relationship between **measured force** and **calculated force**
- Finish **Glove full circuit** and **calibrate** if the calibration method works
- Design the **final data collection** method